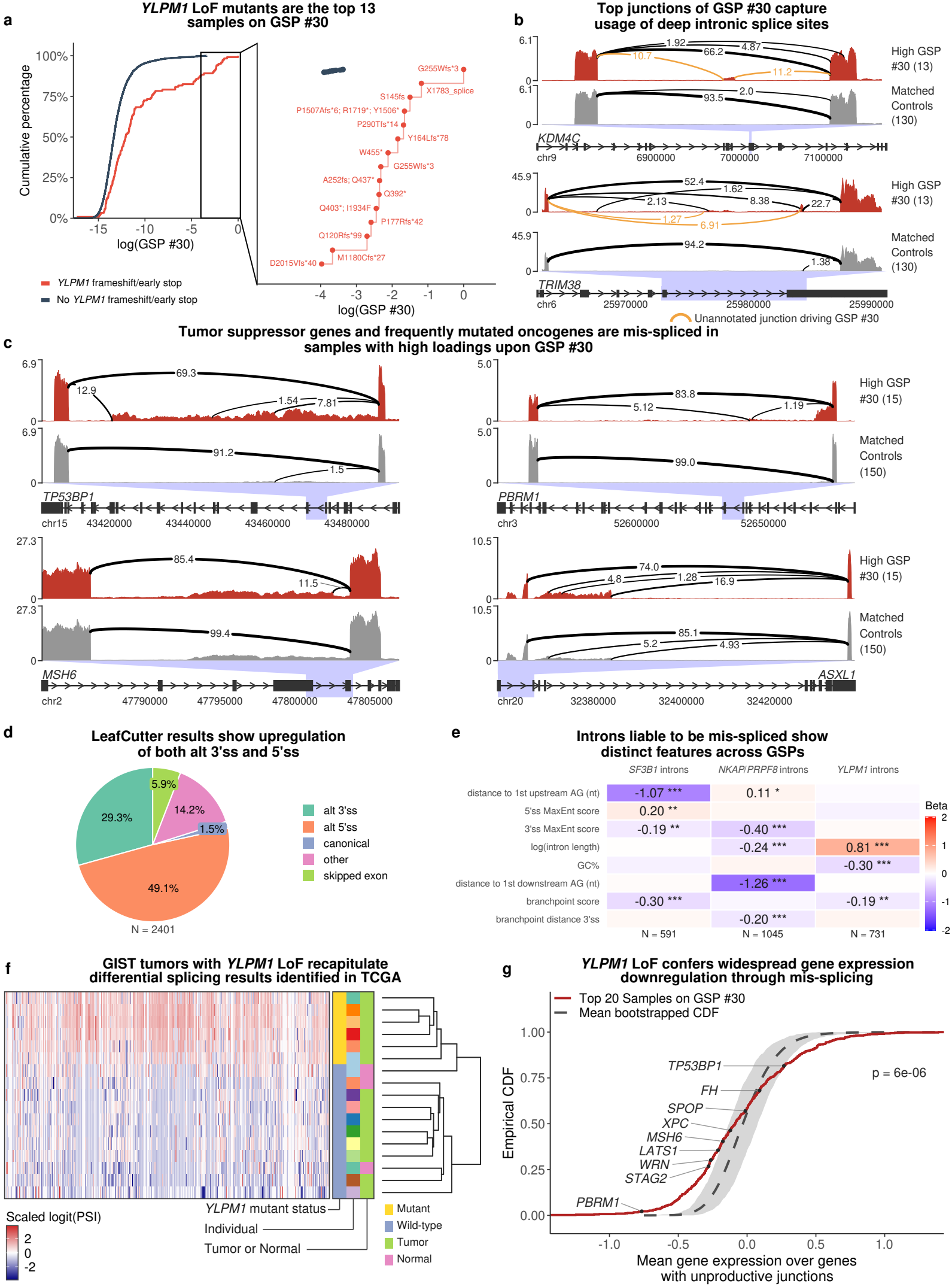

a
YLPM1 LoF mutants are the top 13 samples on GSP #30
Cumulative percentage
log(GSP #30)
YLPM1 frameshift/early stop
No YLPM1 frameshift/early stop
G255Wfs*3
X1783_splice
S145fs
P1507Afs*6; R1719*; Y1506*
P290Tfs*14
Y164Lfs*78
W455*
G255Wfs*3
A252fs; Q437*
Q392*
Q403*; I1934F
P177Rfs*42
Q120Rfs*99
M1180Cfs*27
D2015Vfs*40
b
Top junctions of GSP #30 capture usage of deep intronic splice sites
High GSP #30 (13)
Matched Controls (130)
KDM4C
TRIM38
Unannotated junction driving GSP #30
c
Tumor suppressor genes and frequently mutated oncogenes are mis-spliced in samples with high loadings upon GSP #30
High GSP #30 (15)
Matched Controls (150)
TP53BP1
PBRM1
MSH6
ASXL1
d
LeafCutter results show upregulation of both alt 3'ss and 5'ss
alt 3'ss
alt 5'ss
canonical
other
skipped exon
N = 2401
e
Introns liable to be mis-spliced show distinct features across GSPs
SF3B1 introns
NKAP/PRPF8 introns
YLPM1 introns
distance to 1st upstream AG (nt)
5'ss MaxEnt score
3'ss MaxEnt score
log(intron length)
GC%
distance to 1st downstream AG (nt)
branchpoint score
branchpoint distance 3'ss
Beta
N = 591
N = 1045
N = 731
f
GIST tumors with YLPM1 LoF recapitulate differential splicing results identified in TCGA
Scaled logit(PSI)
YLPM1 mutant status
Individual
Tumor or Normal
Mutant
Wild-type
Tumor
Normal
g
YLPM1 LoF confers widespread gene expression downregulation through mis-splicing
Top 20 Samples on GSP #30
Mean bootstrapped CDF
Empirical CDF
p = 6e-06
TP53BP1
FH
SPOP
XPC
MSH6
LATS1
WRN
STAG2
PBRM1
Mean gene expression over genes with unproductive junctions